How well do charmonium and bottomonium determine the B_c spectrum?

A potential-model test with the $B_c(2S)$ hyperfine difference

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We ask to what extent the spectrum of the B_c meson is fixed by charmonium and bottomonium data alone. Three families of spin-independent potentials—Cornell, logarithmic, and a power law with variable confining exponent α—are refitted, together with the quark masses and the strong coupling, to 27 J-resolved $c\bar{c}$ and $b\bar{b}$ levels from the 2026 Review of Particle Physics, with spin-dependent splittings treated in first-order perturbation theory. Without any B_c input the predictions for $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ agree across families to ± 15 MeV, while P-, D- and higher S-wave levels show 20–150 MeV model dependence that we quantify as a function of α. The $c\bar{c}+b\bar{b}$ data strongly favour a nearly logarithmic potential ($\alpha \lesssim 0.1$). The difference of hyperfine splittings $\Delta_{1S}-\Delta_{2S}$, which CMS and LHCb measure at 29–31 MeV, is predicted at 28–40 MeV and independently selects $\alpha \simeq 0.1$–0.2. With a fixed α_s the absolute $1S$ hyperfine splitting comes out at 76–82 MeV, well above the 64.5 ± 1.7 MeV recently measured by ATLAS; a one-loop running coupling with Λ determined by $c\bar{c}+b\bar{b}$ alone predicts $\Delta_{1S} = 63$–72 MeV and $\Delta_{2S} = 34$–38 MeV, reproducing the ATLAS, CMS and LHCb hyperfine measurements simultaneously without any B_c input. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV above the measured value in every variant. We also give $E1$ and $M1$ radiative widths for the lowest excited B_c states.

1 Introduction

The B_c meson is the only hadron built from two heavy quarks of different flavour. It is therefore both a laboratory for heavy-quark dynamics intermediate between charmonium and bottomonium and a stringent test of any description of the heavy quark–antiquark interaction: a model tuned on $c\bar{c}$ and $b\bar{b}$ must interpolate to $c\bar{b}$ with no additional freedom. Experimentally the field has moved quickly. The ground state is known to 0.3 MeV [1]; CMS [2] and LHCb [3] have observed the $B_c(2S)$ and the $B_c^*(2S)$ doublet; lattice QCD predicted the $B_c^*(1S)$ mass to better than 10 MeV [4]; and in 2026 ATLAS observed the $B_c^*(1S) \to B_c\gamma$ transition directly, measuring the hyperfine splitting $m(B_c^*) - m(B_c) = 64.5 \pm 1.4\,^{+1.0}_{-1.4}$ MeV [5]. Excited P- and D-wave states are within reach of LHCb Run 3.

Non-relativistic potential models [6–8] have long been used to predict the B_c spectrum. Recent work has extended them to higher excitations and radiative decays [9, 10], and has begun to quantify parameter uncertainties with Bayesian methods [9]. A common feature of these studies, however, is that the model parameters are calibrated, at least in part, on the B_c states themselves. This makes agreement with $B_c(1S)$ and $B_c(2S)$ partly circular and obscures the question we wish to address here: *given only the $c\bar{c}$ and $b\bar{b}$ spectra, how well is the B_c spectrum determined, and which of its features are independent of the unknown form of the confining potential?*

We answer this by refitting three potential families to the current $c\bar{c}$ and $b\bar{b}$ data and propagating the fits to $c\bar{b}$ without further adjustment (Secs. 2–4). We then confront the hyperfine sector with three independent measurements: the ATLAS $1S$ splitting, and the $B_c(2S)$–$B_c^*(2S)$ doublet of CMS and LHCb, which determines the *difference* of the $1S$ and $2S$ splittings—a quantity in which the leading systematic uncertainties of the contact interaction cancel (Sec. 5). Radiative widths are given in Sec. 6 and we conclude in Sec. 7.

2 Framework

2.1 Spin-independent spectrum

For a quark of mass m_1 and antiquark of mass m_2 interacting through a central potential $V(r)$, the reduced radial wave function $u_{nl}(r) = rR_{nl}(r)$ obeys

$$u'' = 2\mu\Big[V(r) + \frac{l(l+1)}{2\mu r^2} - E_{nl}\Big]u, \qquad \mu = \frac{m_1 m_2}{m_1+m_2}, \tag{1}$$

in natural units. We integrate Eq. (1) with the Numerov algorithm on a uniform grid ($h = 10^{-3}\,\mathrm{GeV}^{-1}$) and locate eigenvalues by bisection on the number of radial nodes; the results are converged to better than 10^{-6} GeV with respect to h and the integration range. The spin-averaged mass is $M_{nl} = m_1 + m_2 + E_{nl}$.

We consider three families of potential:

$$V_{\mathrm{C}}(r) = -\frac{\kappa}{r} + \frac{r}{a^2}, \tag{2}$$

$$V_{\mathrm{L}}(r) = D + C\ln r, \tag{3}$$

$$V_{\alpha}(r) = V_0 + B\,(r^{\alpha} - 1), \tag{4}$$

with r in GeV^{-1}. Eq. (2) is the Cornell potential [11]; Eq. (3) the Quigg–Rosner logarithmic potential [12]; Eq. (4) is the Martin power law [13] written so that the potential is pinned at $r = 1\,\mathrm{GeV}^{-1}$ ($V_0 = -1.166$ GeV) and the exponent α can be varied without shifting the potential by amounts that the quark masses would otherwise absorb. For $\alpha = 0.1$ it coincides with the Martin potential of Ref. [6]; as $\alpha \to 0$ it tends to Eq. (3) with $C = B\alpha$; $\alpha = 1$ is linear confinement. The additive constants D and V_0 are exactly degenerate with $m_1 + m_2$ and are held fixed.

2.2 Spin-dependent splittings

Fine and hyperfine structure is added in first-order perturbation theory following Eichten and Quigg [6],

$$\Delta = \sum_{k=1}^{4} T_k, \tag{5}$$

$$T_1 = \Big[\frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle}{2m_1^2} + \frac{\langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{2m_2^2}\Big]\tilde{T}_1, \quad T_2 = \frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle + \langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_2,$$
$$T_3 = \frac{\langle \mathbf{s}_1\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_3, \qquad T_4 = \frac{\langle S_{12}\rangle}{m_1 m_2}\tilde{T}_4, \tag{6}$$

with radial coefficients

$$\tilde{T}_1 = -\Big\langle \frac{1}{r}\frac{dV}{dr}\Big\rangle + 2\tilde{T}_2, \qquad \tilde{T}_2 = \frac{4\alpha_s}{3}\langle r^{-3}\rangle,$$
$$\tilde{T}_3 = \frac{8\alpha_s}{9}|R_{n0}(0)|^2, \qquad \tilde{T}_4 = \frac{\alpha_s}{3}\langle r^{-3}\rangle. \tag{7}$$

The full fitted potential is treated as a Lorentz scalar for the spin–orbit term, while the vector one-gluon exchange enters through α_s; this is the prescription of Ref. [6] and keeps the spin-dependent sector free of additional parameters. For $m_1 \neq m_2$ the states with $J = L$ mix; we construct the 2×2 matrix in the $\{^3L_L, {}^1L_L\}$ basis,

$$H = \begin{pmatrix} -\tfrac{1}{2}(a_1+a_2) + 2d & \tfrac{1}{2}(a_1-a_2)\sqrt{L(L+1)} \\ \tfrac{1}{2}(a_1-a_2)\sqrt{L(L+1)} & 0 \end{pmatrix}, \tag{8}$$

with $a_i = \tilde{T}_1/2m_i^2 + \tilde{T}_2/m_1 m_2$ and $d = \tilde{T}_4/m_1 m_2$, and diagonalise it. Its trace and determinant agree with the jj-coupling matrices of Ref. [6], Eqs. (2.21). We label the two $J = L$ eigenstates L^+ and $L^{+\prime}$ in order of mass.

Two treatments of α_s are compared. In the *fixed* variant α_s is a single fitted constant. In the *running* variant we use the one-loop expression $\alpha_s(\mu) = 12\pi/[25\ln(\mu^2/\Lambda^2)]$ ($n_f = 4$) evaluated at a scale set by the quark masses, either $\mu = 2\mu_{\rm red} = 2m_1 m_2/(m_1 + m_2)$ or $\mu = \sqrt{m_1 m_2}$, and fit Λ instead of α_s. The two scale choices coincide for $c\bar{c}$ and $b\bar{b}$ and differ only in how the coupling is interpolated to $c\bar{b}$.

2.3 Radiative transitions

$E1$ and $M1$ widths are computed from the same wave functions [6],

$$\Gamma_{E1} = \frac{4\alpha\langle e_Q\rangle^2}{27}\, k^3\,(2J_f+1)\,\mathcal{S}_{if}\,|\langle f|r|i\rangle|^2, \tag{9}$$
$$\Gamma_{M1} = \frac{16\alpha}{3}\,\mu_Q^2\, k^3\,(2J_f+1)\,|\langle f|j_0(kr/2)|i\rangle|^2, \tag{10}$$

with $\langle e_Q\rangle = (m_b e_c - m_c e_{\bar{b}})/(m_b+m_c)$, $\mu_Q = (m_b e_c - m_c e_{\bar{b}})/4m_c m_b$, $k = (M_i^2 - M_f^2)/2M_i$, and $\mathcal{S}_{if} = 9\max(L_i,L_f)\{\begin{smallmatrix} L_f & J_f & S \\ J_i & L_i & 1\end{smallmatrix}\}^2$, which reduces to $\mathcal{S} = 1$ for ${}^3S_1 \leftrightarrow {}^3P_J$ and $\mathcal{S} = 3$ for ${}^1P_1 \leftrightarrow {}^1S_0$. For spin-mixed states the $E1$ rate is the sum over the $S = 1$ and $S = 0$ channels weighted by the squared mixing amplitudes. On charmonium this reproduces $\Gamma(\chi_{c0,1,2} \to J/\psi\gamma) = 163, 265, 340$ keV against the measured 151, 288, 374 keV [1].

All code is public (see Code and data availability) and every table and figure in this paper is regenerated by a single script.

Table 1: Experimental input: 27 J-resolved levels below open-flavour threshold [1], in MeV. Our radial label n counts excitations from 1, so $\chi_c(1P)$ is $1\,{}^3P_J$. $\Upsilon(4S)$ is excluded.

System	State	Mass
$c\bar{c}$	$1\,{}^1S_0$	2984.1 ± 0.3
$c\bar{c}$	$1\,{}^3S_1$	3096.9 ± 0.0
$c\bar{c}$	$1\,{}^3P_0$	3415.5 ± 0.2
$c\bar{c}$	$1\,{}^3P_1$	3510.7 ± 0.1
$c\bar{c}$	$1\,{}^1P_1$	3525.4 ± 0.1
$c\bar{c}$	$1\,{}^3P_2$	3556.2 ± 0.1
$c\bar{c}$	$2\,{}^1S_0$	3637.8 ± 0.6
$c\bar{c}$	$2\,{}^3S_1$	3686.1 ± 0.0
$c\bar{c}$	$1\,{}^3D_1$	3773.7 ± 0.7
$c\bar{c}$	$1\,{}^3D_2$	3823.5 ± 0.3
$c\bar{c}$	$1\,{}^3D_3$	3842.7 ± 0.2
$b\bar{b}$	$1\,{}^1S_0$	9398.7 ± 2.0
$b\bar{b}$	$1\,{}^3S_1$	9460.4 ± 0.1
$b\bar{b}$	$1\,{}^3P_0$	9859.4 ± 0.5
$b\bar{b}$	$1\,{}^3P_1$	9892.8 ± 0.4
$b\bar{b}$	$1\,{}^1P_1$	9899.3 ± 0.8
$b\bar{b}$	$1\,{}^3P_2$	9912.2 ± 0.4
$b\bar{b}$	$2\,{}^1S_0$	9999.0 ± 3.5
$b\bar{b}$	$2\,{}^3S_1$	10023.4 ± 0.5
$b\bar{b}$	$1\,{}^3D_2$	10163.7 ± 1.4
$b\bar{b}$	$2\,{}^3P_0$	10232.5 ± 0.6
$b\bar{b}$	$2\,{}^3P_1$	10255.5 ± 0.6
$b\bar{b}$	$2\,{}^1P_1$	10259.8 ± 1.2
$b\bar{b}$	$2\,{}^3P_2$	10268.6 ± 0.6
$b\bar{b}$	$3\,{}^3S_1$	10355.1 ± 0.5
$b\bar{b}$	$3\,{}^3P_1$	10513.4 ± 0.7
$b\bar{b}$	$3\,{}^3P_2$	10524.0 ± 0.8

3 Fit to charmonium and bottomonium

Table 1 lists the input. We fit each family's potential parameters, m_c, m_b and α_s (or Λ) by least squares to all 27 levels, using $\sigma^2 = \sigma_{\rm exp}^2 + \sigma_{\rm th}^2$ with a theory floor $\sigma_{\rm th} = 10$ MeV so that the keV-precision J/ψ and $\psi(2S)$ masses do not dominate a model whose intrinsic accuracy is tens of MeV. Parameter uncertainties are taken from the covariance of the fit and later propagated linearly to predictions.

The results are in Table 2 and Fig. 1. Residuals are typically 10–30 MeV and are largest for the charmonium D waves (all families miss $\psi_3(3842)$ by ~ 40 MeV), which lie at the $D\bar{D}$ threshold where coupled-channel effects are expected. Compared with the 1994 parameters [6], the logarithmic and Martin families change little except for α_s, which rises from 0.37–0.43 to 0.45–0.46; the Cornell quark masses, by contrast, drop by 0.4–0.5 GeV, confirming that the

Table 2: Fitted parameters and χ^2 per degree of freedom for a 10 MeV theory floor. Units: B, C, m_c, m_b, Λ in GeV; a in GeV^{-1}. For the Martin family B, m_c and m_b are strongly correlated (see text); the individual errors are large but predicted masses are not.

Family	χ^2/dof	Parameters
Martin	84.7/22	$B = 7.19 \pm 0.45$, $\alpha = 0.096(6)$, $m_c = 1.63 \pm 0.25$, $m_b = 5.02 \pm 0.23$, $\alpha_s = 0.457(26)$
Log	73.3/23	$C = 0.737(5)$, $m_c = 1.509(4)$, $m_b = 4.908(2)$, $\alpha_s = 0.448(22)$
Cornell	133.7/22	$\kappa = 0.456(10)$, $a = 2.328(20)$, $m_c = 1.334(9)$, $m_b = 4.752(7)$, $\alpha_s = 0.373(19)$
Martin, running α_s	69.8/22	$B = 6.84 \pm 0.49$, $\alpha = 0.101(7)$, $m_c = 1.82 \pm 0.27$, $m_b = 5.20 \pm 0.26$, $\Lambda = 0.429(94)$
Log, running α_s	58.8/23	$C = 0.740(5)$, $m_c = 1.509(4)$, $m_b = 4.905(2)$, $\Lambda = 0.332(25)$

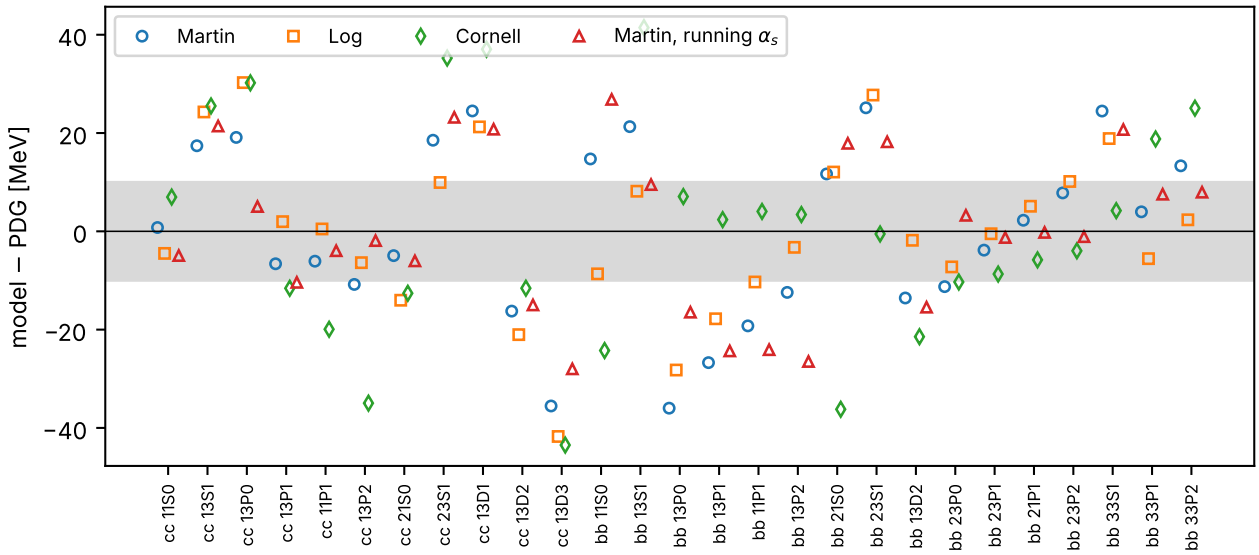


Figure 1: Residuals (model − PDG) of the fits for the three potential families and the running-coupling Martin fit. The grey band is the 10 MeV theory floor.

original Cornell masses were tuned to a different mass convention. The running coupling improves the fit in both families it was applied to (χ^2: 85 → 70 for Martin, 74 → 59 for logarithmic) because it allows the $c\bar{c}$ and $b\bar{b}$ hyperfine splittings, 113 and 62 MeV, to be reproduced simultaneously; the fitted $\Lambda = 0.33$–0.43 GeV corresponds to $\alpha_s(m_c) \simeq 0.50$ and $\alpha_s(m_b) \simeq 0.28$–$0.30$.

For the Martin family the parameters B, m_c and m_b carry large formal errors because $B(r^\alpha - 1) \simeq B\alpha \ln r$ for small α, so B and the masses are nearly degenerate. The combination $m_c + m_b$ that enters B_c is nevertheless well constrained, and the propagated uncertainties on predicted masses are $\lesssim 15$ MeV.

To probe the shape of the confining potential we repeat the fit of Eq. (4) at fixed $\alpha \in \{0.05, 0.1, 0.2, 0.35, 0.5, 0.75, 1\}$ (Table 3, Fig. 2). The χ^2 rises monotonically from 73 at $\alpha = 0.05$ to 986 at $\alpha = 1$: the $c\bar{c}$ and $b\bar{b}$ spectra together strongly prefer a nearly logarithmic potential, consistent with the classic observation that the $2S$–$1S$ spacing is almost flavour independent [12]. A free fit of α returns 0.096(6).

4 The B_c spectrum

Table 4 gives the B_c spectrum through $3S$, $2P$ and $1D$ predicted by the three families, and Fig. 3 shows the dependence on α across the scan. Three features stand out.

First, the S-wave levels are remarkably stable. $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ vary by ± 14, ± 17 and ± 6 MeV across the three families and by less than 70 MeV over the whole α scan, even though χ^2 changes by an order of magnitude. The ground state is reproduced at 6276 ± 14 MeV against the measured 6274.5 MeV without any B_c input.

Second, the excited P, D and $3S$ levels are not stable: the $1P$ centroid moves by 80 MeV and the $3\,^3S_1$ by 150 MeV between $\alpha = 0$ and $\alpha = 1$, and the Cornell family—which the $c\bar{c} + b\bar{b}$ fit disfavours—sits 30–60 MeV above the others for the $2P$ and $3S$. These are the states for which future measurements will discriminate between confinement models; restricted to the $\alpha \lesssim 0.2$ region preferred by $c\bar{c} + b\bar{b}$, the $1P$ centroid is 6737–6768 MeV and the $1D$ centroid 7023–7043 MeV.

Third, the spin-averaged $B_c(2S)$ is predicted 11–24 MeV *above* the measured 6871.2 MeV in every one of the thirteen fits performed in this work (Tables 4 and 3), with a propagated uncertainty of only 2–

Table 3: Fits at fixed confining power α and the resulting B_c levels (MeV). Centroids are spin-weighted averages of the fine-structure multiplets. The last column is the predicted $\Delta_{1S}-\Delta_{2S}$; experiment gives 29.1 ± 1.7 MeV [2]. $\alpha=0$ denotes the logarithmic fit.

α	χ^2/dof	α_s	$B_c(1S)$	$B_c^*(1S)$	$B_c(2S)$	$1P$	$1D$	$3\,^3S_1$	$\Delta_{1S}-\Delta_{2S}$
0	73/23	0.448	6266	6349	6889	6768	7043	7253	40.3
0.05	72/23	0.461	6257	6337	6878	6752	7030	7254	37.2
0.1	85/22	0.457	6267	6343	6886	6753	7034	7270	33.8
0.2	136/23	0.465	6269	6338	6882	6737	7023	7285	27.3
0.35	254/23	0.465	6280	6340	6886	6724	7014	7312	19.4
0.5	404/23	0.462	6292	6344	6889	6713	7006	7336	13.1
0.75	689/23	0.457	6309	6351	6893	6698	6990	7371	5.3
1	986/23	0.451	6325	6360	6897	6686	6976	7400	0.0

Table 4: B_c spectrum (MeV) predicted from the $c\bar{c}+b\bar{b}$ fits with fixed α_s. Parenthesised errors are the propagated fit uncertainties; the model spread is half the range over the three families. The comparison columns are Eichten and Quigg (Buchmüller–Tye potential) [6] and PDG 2026 [1]. Our $1P$ and $1D$ correspond to the $2P$ and $3D$ of the principal-quantum-number labelling.

State	Martin	Log	Cornell	mean $\pm$ spread	EQ94	PDG
$1\,^1S_0$	6267(11)	6266(4)	6295(4)	6276 ± 14	6264	6274.5 ± 0.32
$1\,^3S_1$	6343(10)	6349(4)	6376(4)	6356 ± 17	6337	6339.0 ± 1.7
$1\,^3P_0$	6692(15)	6711(4)	6724(4)	6709 ± 16	6700	–
1^+	6735(13)	6753(3)	6753(3)	6747 ± 9	6730	–
$1^{+\prime}$	6756(10)	6770(3)	6759(2)	6762 ± 7	6736	–
$1\,^3P_2$	6774(10)	6787(3)	6770(3)	6777 ± 9	6747	–
$2\,^1S_0$	6886(11)	6889(2)	6877(3)	6884 ± 6	6856	6871.2 ± 1.0
$2\,^3S_1$	6928(10)	6931(2)	6930(3)	6930 ± 1	6899	–
$1\,^3D_1$	7029(14)	7040(3)	7036(3)	7035 ± 5	7012	–
$1D_2$	7029(9)	7037(3)	7022(3)	7029 ± 7	7009	–
$1D_2'$	7040(14)	7050(3)	7044(3)	7044 ± 5	7012	–
$1\,^3D_3$	7035(9)	7043(3)	7027(3)	7035 ± 8	7005	–
$2\,^3P_0$	7102(13)	7103(4)	7148(5)	7118 ± 23	7108	–
1^+	7132(12)	7131(3)	7176(4)	7146 ± 23	7135	–
$1^{+\prime}$	7151(10)	7147(3)	7185(4)	7161 ± 19	7142	–
$2\,^3P_2$	7164(10)	7158(3)	7196(4)	7173 ± 19	7153	–
$3\,^1S_0$	7240(11)	7225(4)	7290(6)	7251 ± 32	7244	–
$3\,^3S_1$	7270(10)	7253(4)	7335(6)	7286 ± 41	7280	–

11 MeV. Since the $c\bar{c}$ and $b\bar{b}$ $2S$ levels are themselves reproduced to $\lesssim 25$ MeV with residuals of both signs, this is a genuine interpolation failure at the few-MeV level rather than a fit artefact, and is the most robust discrepancy we find. We return to it in Sec. 7.

Our fixed-α_s $1P$ and $1D$ levels lie 5–30 MeV above those of Ref. [6], whose Buchmüller–Tye potential is softer at the relevant distances; the ordering within each multiplet and the mixing of the $J=L$ states (mixing angle $\simeq 23^\circ$ for the $1P$) are the same.

5 The hyperfine test

In both the CMS and LHCb analyses the $B_c^*(2S)$ is reconstructed through $B_c^*(2S)\to B_c^*(1S)\pi\pi$, $B_c^*(1S)\to B_c\gamma$ with the soft photon unobserved, so the observed peak sits at $M(B_c^*(2S))-\Delta_{1S}$, where $\Delta_{nS}\equiv M(n^3S_1)-M(n^1S_0)$. The measured separation of the two peaks is therefore

$$M(B_c(2S))-M_{\rm peak}=\Delta_{1S}-\Delta_{2S}, \qquad (11)$$

$=29.1\pm1.5\pm0.7$ MeV (CMS [2]) and 30.9 MeV (LHCb [3]). This is a quantity in which potential models are unusually trustworthy. From Eq. (7),

$$\Delta_{1S}-\Delta_{2S}=\frac{8\alpha_s}{9m_cm_b}\big[|R_{10}(0)|^2-|R_{20}(0)|^2\big], \qquad (12)$$

so the overall normalisation uncertainties of the contact interaction—the value of α_s, relativistic corrections to $|R(0)|^2$, the choice of quark masses—enter as a common factor, while the shape of the potential enters through the ratio $|R_{20}(0)|^2/|R_{10}(0)|^2$.

Table 5 and Fig. 4 show that the three fixed-α_s families predict $\Delta_{1S}-\Delta_{2S}=28$–40 MeV, bracketing the measurement, and that across the α scan (Fig. 2, Table 3) the prediction falls smoothly from 37 MeV at $\alpha=0.05$ to zero at $\alpha=1$, where $|R_{n0}(0)|^2$ is independent of n. The measurement therefore selects $\alpha\simeq0.1$–

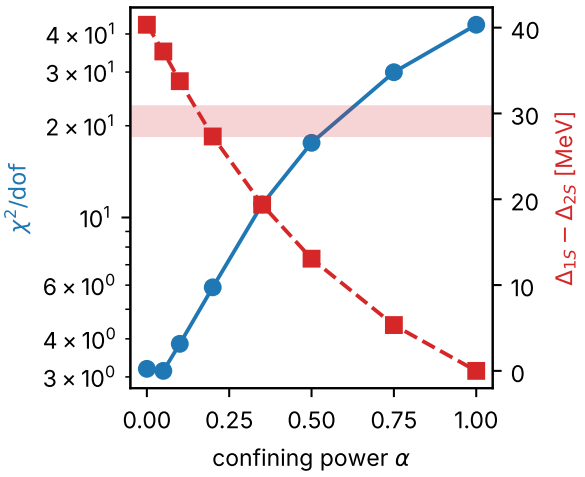


Figure 2: Left axis: χ^2/dof of the $c\bar{c}+b\bar{b}$ fit as a function of the confining power α in Eq. (4). Right axis: the predicted hyperfine difference $\Delta_{1S}-\Delta_{2S}$ of B_c; the band is the CMS measurement [2].

Table 5: Hyperfine splittings (MeV) of B_c predicted from $c\bar{c}+b\bar{b}$, for fixed and running α_s. Experiment: $\Delta_{1S}=64.5\pm1.7$ and $M(B_c^*)=6339.0\pm1.7$ (ATLAS [5]); $\Delta_{1S}-\Delta_{2S}=29.1\pm1.7$ (CMS [2]) and 30.9 (LHCb [3], statistical error ≈1.5); hence $\Delta_{2S}=35.4\pm2.4$. Lattice QCD (2010): $M(B_c^*)=6330\pm9$ [4].

Family	Δ_{1S}	Δ_{2S}	$\Delta_{1S}-\Delta_{2S}$	$M(B_c^*)$
Martin	75.8	42.0	33.8	6343
Log	82.4	42.0	40.3	6349
Cornell	81.1	53.0	28.1	6376
Martin, running ($\sqrt{m_c m_b}$)	62.8	35.0	27.8	6331
Martin, running (2μ)	67.3	37.5	29.8	6332
Log, running ($\sqrt{m_c m_b}$)	66.4	33.9	32.5	6342
Log, running (2μ)	72.0	36.8	35.3	6343

Table 6: $E1$ widths above 10 keV and $M1$ widths of the lowest B_c states, for the Martin fit with fixed α_s. k is the photon energy. Spin-mixed states are treated as described in Sec. 2.

Transition	Type	k [MeV]	Γ [keV]
$1^{+\prime}\to1\,^1S_0\,\gamma$	E1	472	135.3
$1\,^3P_2\to1\,^3S_1\,\gamma$	E1	417	110.8
$1\,^3D_1\to1\,^3P_0\,\gamma$	E1	329	94.6
$1\,^3D_3\to1\,^3P_2\,\gamma$	E1	256	80.6
$1^+\to1\,^3S_1\,\gamma$	E1	381	71.4
$1^{+\prime}\to2\,^1S_0\,\gamma$	E1	260	65.7
$1\,^3P_0\to1\,^3S_1\,\gamma$	E1	340	60.1
$2\,^3P_2\to2\,^3S_1\,\gamma$	E1	232	57.1
$1D_2'\to1^+\,\gamma$	E1	298	52.5
$1D_2\to1^{+\prime}\,\gamma$	E1	267	46.7
$1D_2'\to1^{+\prime}\,\gamma$	E1	278	46.2
$1D_2\to1^+\,\gamma$	E1	287	42.4
$1\,^3D_1\to1^+\,\gamma$	E1	288	40.2
$1^+\to2\,^3S_1\,\gamma$	E1	202	30.4
$2\,^3P_0\to2\,^3S_1\,\gamma$	E1	172	23.1
$1^+\to1\,^1S_0\,\gamma$	E1	452	21.5
$1^{+\prime}\to1\,^1S_0\,\gamma$	E1	829	18.7
$2\,^3P_2\to1\,^3S_1\,\gamma$	E1	774	18.6
$1^{+\prime}\to1\,^3S_1\,\gamma$	E1	401	15.0
$2\,^3S_1\to1\,^3P_2\,\gamma$	E1	152	15.0
$2\,^3P_0\to1\,^3S_1\,\gamma$	E1	718	14.9
$2\,^3S_1\to1^+\,\gamma$	E1	190	14.7
$2\,^1S_0\to1^{+\prime}\,\gamma$	E1	128	13.7
$1^+\to1\,^3S_1\,\gamma$	E1	746	13.6
$2\,^3P_2\to1\,^3D_3\,\gamma$	E1	128	12.8
$1^+\to2\,^1S_0\,\gamma$	E1	242	11.9
$1D_2'\to1\,^3P_2\,\gamma$	E1	261	11.5
$2\,^3S_1\to1\,^3P_0\,\gamma$	E1	232	10.5
$1\,^3S_1\to1\,^1S_0\,\gamma$	M1	75	0.1
$2\,^3S_1\to1\,^1S_0\,\gamma$	M1	629	0.1
$2\,^1S_0\to1\,^3S_1\,\gamma$	M1	522	0.1
$2\,^3S_1\to2\,^1S_0\,\gamma$	M1	42	0.0

0.2, in agreement with—and independent of—the preference of the $c\bar{c}+b\bar{b}$ spectrum itself. Two unrelated observables point to the same confining potential.

The absolute splitting is a sharper test still, now that ATLAS has measured it directly: $\Delta_{1S}=64.5\pm1.7$ MeV [5], some 9 MeV (one standard deviation) above the 2010 lattice prediction [4]. Combined with the CMS difference this fixes $\Delta_{2S}=35.4\pm2.4$ MeV, so the two lowest hyperfine splittings of B_c are both known to a few MeV. Figure 5 and Table 5 compare them with the predictions. With fixed α_s all three families give $\Delta_{1S}=76$–82 MeV and $\Delta_{2S}=42$–53 MeV, 10–20 MeV too high, even though each reproduces the $c\bar{c}$ and $b\bar{b}$ $1S$ splittings to $\lesssim20$ MeV: a single coupling cannot interpolate the contact term $\propto\alpha_s|R(0)|^2/m_1m_2$ between flavours. With the one-loop running coupling the Martin family gives $\Delta_{1S}=63$–67 MeV, $\Delta_{2S}=35$–38 MeV and $M(B_c^*)=6331$–6332 MeV, and the logarithmic family $\Delta_{1S}=67$–72 MeV, $\Delta_{2S}=34$–37 MeV—both within or close to the ATLAS and CMS uncertainties. We stress that Λ in these fits is determined by the $c\bar{c}$ and $b\bar{b}$ hyperfine splittings alone, and that neither a fixed coupling nor any adjustment of the spin-independent potential can reproduce the measured values. A one-parameter running coupling fitted to equal-mass quarkonia therefore predicts the unequal-mass hyperfine sector of B_c—the absolute $1S$ splitting, the $2S$ splitting and their difference—without B_c input. The same fits place the lattice value [4] at the low edge of the allowed range; an updated lattice determination of $m(B_c^*)$ with current precision would be valuable.

6 Radiative transitions

Table 6 lists the $E1$ and $M1$ widths of the $1P$, $1D$ and $2P$ states. The $1P\to1S$ widths of 60–135 keV agree in magnitude with Refs. [6, 10]; the spin mixing of the 1^+ states redistributes the rate so that the heavier $1^{+\prime}$ decays mainly to $B_c\gamma$ (135 keV) and the lighter 1^+ mainly to $B_c^*\gamma$ (71 keV). The $1D\to1P$ widths of 40–95 keV make the cascade $1D\to1P\gamma\to1S\gamma\gamma$

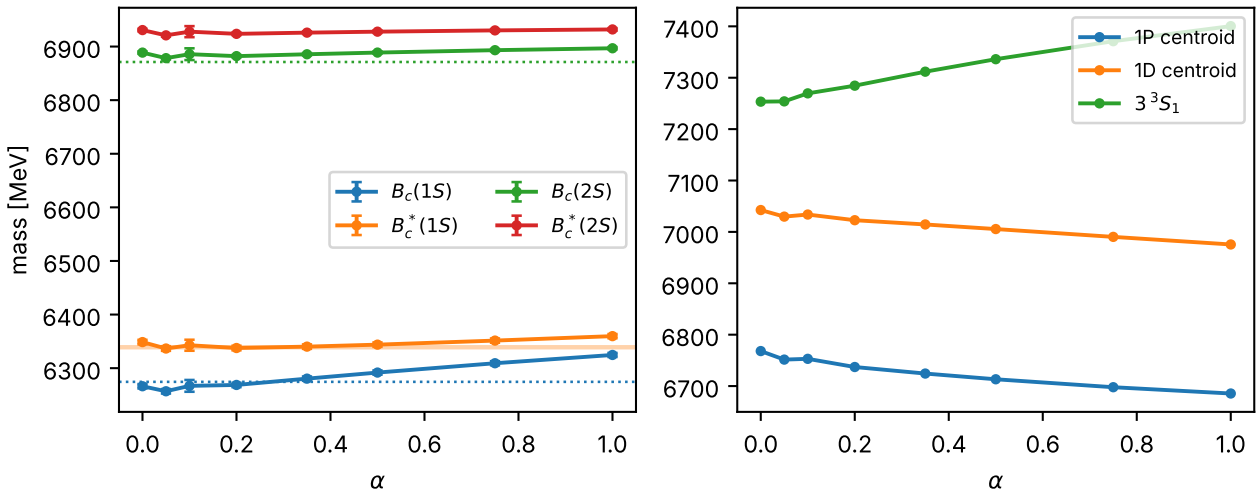


Figure 3: B_c levels as a function of the confining power α of the potential fitted to $c\bar{c} + b\bar{b}$. Left: S-wave states; dotted lines are the measured $B_c(1S)$ and $B_c(2S)$, the band is the ATLAS $B_c^*(1S)$ [5]. Right: $1P$ and $1D$ centroids and the $3\,^3S_1$.

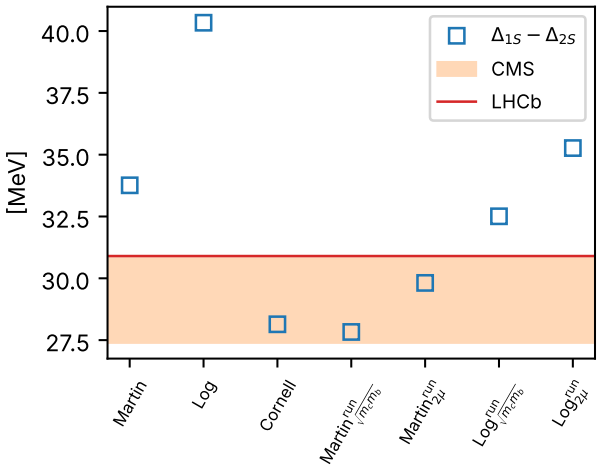


Figure 4: Predicted $\Delta_{1S} - \Delta_{2S}$ for each fit, compared with CMS (band) and LHCb (line).

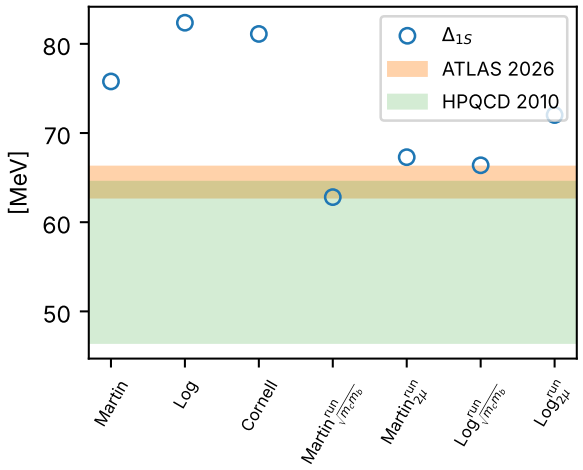


Figure 5: Predicted absolute $1S$ hyperfine splitting Δ_{1S} for each fit, compared with the ATLAS measurement [5] (narrow band) and the 2010 lattice prediction [4] (wide band).

the natural route to the first D-wave B_c state. The $M1$ rates are $\lesssim 0.1$ keV; the $B_c^*(1S) \to B_c\gamma$ photon ($k \simeq 60$–75 MeV) remains the principal obstacle to a direct B_c^* mass measurement. We have not included the model dependence of these widths, which is dominated by the photon energy ($\propto k^3$) and is therefore largest for transitions involving the excited levels of Sec. 4.

7 Conclusions

Fitting three families of potentials to the 2026 charmonium and bottomonium spectra and propagating them to $c\bar{b}$ without further input, we find:

1. The $c\bar{c} + b\bar{b}$ data prefer a nearly logarithmic confining potential, $\alpha \lesssim 0.1$, and the S-wave B_c levels are determined to ± 15 MeV independently of the potential; the P, D and $3S$ levels are not, and are the states to measure.

2. The hyperfine difference $\Delta_{1S} - \Delta_{2S}$ measured by CMS and LHCb is predicted correctly from $c\bar{c}+b\bar{b}$ and independently selects $\alpha \simeq 0.1$–0.2.

3. A fixed α_s overpredicts the $B_c^*(1S)$ hyperfine splitting measured by ATLAS by 10–20 MeV; a one-loop running coupling, with Λ fixed by $c\bar{c}$ and $b\bar{b}$, predicts $\Delta_{1S} = 63$–72 MeV and $\Delta_{2S} = 34$–38 MeV, in agreement with ATLAS, CMS and LHCb simultaneously.

4. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV too high in every variant.

The last point deserves comment. The $2S$ discrepancy survives changes of confining potential, of α_s treatment and of theory floor, and is several times the propagated uncertainty. It is the first level at which the first-order, non-relativistic treatment used here appears to fail for B_c while succeeding for $c\bar{c}$ and $b\bar{b}$, and we suggest that it reflects either relativistic corrections that do not interpolate linearly in $1/m$, or the proximity of the $2S$ to the BD threshold. A relativised treatment [7] or a coupled-channel calculation with the $c\bar{c} + b\bar{b}$-determined potential of this work would settle the question.

Acknowledgments

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Code and data availability

The complete code, the fitted parameter sets and the scripts that regenerate every table and figure in this paper are available at `https://github.com/apoppong24/bc-quarkonium`; the experimental inputs are listed in Table 1.

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